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Goodness-of-fit tests for βARMA hydrological time series modeling

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Abstract
We address the issue of performing portmanteau testing inference using time series data that assume values in the standard unit interval. The motivation involves modeling the time series dynamics of the proportion of stocked hydroelectric energy in the South of Brazil. Our focus lies in the class of beta autoregressive moving average (βARMA) models. In particular, we wish to test the goodness-of-fit of such models. We consider several testing criteria that have been proposed for Gaussian time series models and introduce two new tests. We derive the asymptotic null distribution of the two proposed test statistics in two different scenarios, namely, when the tests are applied to an observed time series and when they are applied to the residuals from a fitted βARMA model. It is worth noticing that our results imply the asymptotic validity of standard portmanteau tests in the class of βARMA models that are, under the null hypothesis, asymptotically equivalent to our test statistics. We use Monte Carlo simulation to assess the relative merits of the different portmanteau tests when used with fitted βARMA models. The simulation results we present show that the new tests are typically more powerful than a well-known test whose test statistic is also based on residual partial autocorrelations. Overall, the tests we propose perform quite well. Finally, we model the dynamics of the proportion of stocked hydroelectric energy in Brazil. The results show that the βARMA model outperforms three alternative models and an exponential smoothing algorithm.

KEYWORDS
bootstrap, hydrological data, Monte Carlo simulation, portmanteau test, β ARMA

1 | INTRODUCTION

The beta regression model was introduced by Ferrari and Cribari-Neto (2004) for modeling dependent variables that assume values in the standard unit interval (see also Cribari-Neto & Zeileis, 2010). On the basis of that model, Rocha and Cribari-Neto (2009) introduced the beta autoregressive moving average (βARMA) model, which is a dynamic model tailored for time series that assume values in $(0, 1)$, such as rates and proportions. Doubly bounded random variables are typically asymmetrically distributed, and inferences based on the Gaussian assumption may be quite inaccurate. In the βARMA model, the variable of interest (y) is assumed to follow the beta law, its mean being impacted by a set of covariates and subject to autoregressive and moving average dynamics. Novel features of the model are that it requires no data transformation and that βARMA fitted values and out-of-sample forecasts will never fall outside the standard unit interval.

Diagnostic analysis is of paramount importance in time series modeling. It is performed after the model has been identified and fitted. Different model validation strategies can be used. Perhaps the most commonly used validation strategy involves portmanteau testing inference. Such tests are based on statistics that use residual autocorrelations. They seek to detect any existing serial correlation in the residuals obtained from the fitted model.

Since the seminal article by Box and Pierce (1970), considerable attention has been devoted to tests that use residual autocorrelations to assess goodness-of-fit. Ljung and Box (1978) showed that a simple modification to the test statistic proposed by Box and Pierce considerably reduces the distribution location bias and improves the quality of the asymptotic approximation used in the test. Their statistic is widely used by practitioners (see, e.g., Chiogna & Gaetan, 2005). Monti (1994) proposed to base portmanteau testing inference on a test statistic that uses residual partial autocorrelations rather than residual autocorrelations. Dufour and Roy (1986) introduced a nonparametric portmanteau test based on rank autocorrelations. Their test is particularly useful when the underlying distribution of the time series is unknown, and it tends to deliver accurate inferences under non-normality. Additional tests based on transformed sample autocorrelations were proposed by Kwan and Sim (1996a, 1996b). Peña and Rodriguez (2002) introduced a test based on the determinant of the residual autocorrelation matrix. They approximated the test statistic null distribution using the gamma distribution. Such an approximation can be poor in some situations. To circumvent such a shortcoming, Lin and McLeod (2006) proposed using bootstrap resampling to estimate the test statistic null distribution.

All aforementioned portmanteau tests were developed for standard ARMA models, that is, for models used with variables that assume values in the real line. How do such tests perform when used with βARMA models? To the best of our knowledge, this question remains unanswered. The contribution of our paper to the literature is twofold. First, we investigate the accuracy of portmanteau testing inference in the class of βARMA models. We show that all tests are nearly free of size distortions when coupled with bootstrap resampling. By size distortion, we mean the difference between exact and nominal Type I error frequencies. Second, we introduce two new portmanteau test statistics that are based on residual partial autocorrelations. We base the test statistics on partial autocorrelations because when the order of moving average dynamics is underestimated, the sum of squared partial autocorrelations is likely to be larger than that of squared autocorrelations. We then derive the asymptotic null distribution of the two test statistics in two different scenarios, namely, when the tests are applied to an observed time series and when they are applied to the residuals from a fitted βARMA model. A novel aspect of our proofs is that they also hold for the test statistics that were proposed in the Gaussian time series literature, and that can be shown to be asymptotically equivalent to our test statistics under the null hypothesis. Hence, it follows that some portmanteau tests that were proposed for Gaussian ARMA models can also be used with βARMA models.

The simulation evidence we present shows that the two tests we propose are particularly powerful. In some scenarios of our numerical experiments, such tests were more powerful (i.e., more capable of detecting model misspecification) than all competing tests.

Our motivation for this paper lies in a hydrological empirical problem: We wish to model the time series dynamics of the proportion of stocked hydroelectric energy in the South of Brazil. The data range from January 2001 to October 2016. We produce out-of-sample forecasts using a fitted βARMA model, another dynamic model tailored for time series that assume values in the standard unit interval, a Gaussian ARMA model, a Gaussian autoregressive (AR) model, and an exponential smoothing algorithm. It is shown that βARMA yields the most accurate short-term forecasts. Indeed, the six-period-ahead mean absolute forecasting error obtained with the βARMA model is over 16% smaller than that of the second-best performing method. Prior to using the fitted βARMA model for forecasting, we validate it on the basis of portmanteau testing inference. We shall return to this application in Section 6.

This paper unfolds as follows. Section 2 presents the βARMA model and its main properties. In Section 3, we review some portmanteau tests for model adequacy that have been used in the literature. Two new tests are introduced in Section 4. The asymptotic null distribution of the proposed test statistics is derived under two different settings, namely, when the test statistics are computed using an observed time series and when they are computed using βARMA residuals. Monte Carlo simulation evidence is presented in Section 5. An empirical hydrological application is presented and discussed in Section 6. Finally, Section 7 contains some concluding remarks.

2 | THE MODEL

The βARMA (Rocha & Cribari-Neto, 2009) model is a dynamic model based on the class of beta regression models (Ferrari & Cribari-Neto, 2004). It is useful for dealing with time series data that assume values in the standard unit interval, (0, 1), such as rates and proportions. The model includes autoregressive and moving average dynamics as well as a set of regressors. It accommodates distributional asymmetries and nonconstant dispersion. Unlike the standard ARMA model, fitted values and out-of-sample forecasts produced using βARMA are guaranteed to lie inside the standard unit interval.

Let $\boldsymbol{y} = (y_1, \dots, y_n)'$ be a vector containing n random variables, where each y_t, for $t = 1, \dots, n$, given the previous information set $\mathcal{F}_{t-1}$, follows the beta law, as parameterized in Ferrari and Cribari-Neto (2004), with conditional mean μ_t and precision parameter ϕ. The conditional density of y_t given $\mathcal{F}_{t-1}$ is

$$f(y_t|\mathcal{F}_{t-1}) = \frac{\Gamma(\phi)}{\Gamma(\mu_t\phi)\Gamma((1-\mu_t)\phi)} y_t^{\mu_t\phi-1}(1-y_t)^{(1-\mu_t)\phi-1}, \quad 0 < y_t < 1, \tag{1}$$

where $0 < \mu_t < 1$, $\phi > 0$ and $\Gamma(\cdot)$ is the gamma function. The conditional mean and the conditional variance of y_t are, respectively, $\mathbb{E}(y_t|\mathcal{F}_{t-1}) = \mu_t$ and $\mathrm{var}(y_t|\mathcal{F}_{t-1}) = \mu_t(1-\mu_t)/(1+\phi)$. Note that μ_t is the mean of y_t and ϕ is a precision parameter, in the sense that, for a fixed μ_t, the variance of y_t decreases as ϕ increases.

By assuming that the variable of interest follows the above beta law, Rocha and Cribari-Neto (2009) proposed the following βARMA(p, q) model:

$$g(\mu_t) = \alpha + \boldsymbol{x}_t'\boldsymbol{\beta} + \sum_{i=1}^{p} \varphi_i \left\{ g(y_{t-i}) - \boldsymbol{x}_{t-i}'\boldsymbol{\beta} \right\} + \sum_{j=1}^{q} \theta_j r_{t-j}, \tag{2}$$

where $\boldsymbol{x}_t' \in \mathbb{R}^s$ is a set of nonrandom covariates at time t, $\boldsymbol{\beta} = (\beta_1, \dots, \beta_s)' \in \mathbb{R}^s$ is a vector of coefficients related to the covariates, and $g : (0, 1) \to \mathbb{R}$ is a twice-differentiable strictly monotonic link function. Here, $\alpha \in \mathbb{R}$ is a scalar parameter, and $p, q \in \mathbb{N}$ are, respectively, the autoregressive and moving average orders. Additionally, r_t is the error term, and $\boldsymbol{\varphi} = (\varphi_1, \dots, \varphi_p)'$ and $\boldsymbol{\theta} = (\theta_1, \dots, \theta_q)'$ are the autoregressive and moving average vectors of parameters, respectively. The covariates are only required to be nonrandom and to satisfy (2) for the model to be well defined. Typically, the covariates are included in the model when some deterministic behavior (such as a cyclic or a seasonal component) is present in the data dynamics (see Section 4.2 for details).

The formulation of model (2) is similar to that in Benjamin, Rigby, and Stasinopoulos (2003), who introduced the class of generalized autoregressive moving average (GARMA) models by extending previous results by Li (1994) and Zeger and Qaqish (1988) on non-Gaussian time series modeling. In both classes of models, the error term (r_t) is defined in residual fashion; that is, the errors do not drive the stochastic process like innovations in standard ARMA models. The Gaussian ARMA process is driven by the realization of a white noise error from a normal distribution. There is no white noise error disturbance in βARMA and GARMA models. The time series realizations come from a conditional distribution, and the moving average error, r_t, is defined as the difference between an observed quantity (y_t or $g(y_t)$) and the corresponding model-based quantity (μ_t or $g(\mu_t)$, respectively). Notice that only past values of r_t are included in model (2). The two standard formulations for r_t are (a) error on the original scale: $y_t - \mu_t$ and (b) error on the predictor scale: $g(y_t) - \eta_t$, where $\eta_t = g(\mu_t)$. In what follows, we shall consider the latter. The error term r_t is $\mathcal{F}_{t-1}$-measurable. It is noteworthy that the error in the original scale ($r_t = y_t - \mu_t$) follows a martingale difference which implies that r_t has unconditional mean zero and is unconditionally uncorrelated. Notice that $\mathbb{E}(y_t|\mathcal{F}_{t-1}) = \mu_t$, which implies that $\mathbb{E}(r_t|\mathcal{F}_{t-1}) = 0$ and, hence, $\mathbb{E}(r_t) = 0$. Additionally, for $h \geq 1$, $\mathrm{cov}(r_t, r_{t+h}) = \mathbb{E}(r_t r_{t+h}) = \mathbb{E}(\mathbb{E}(r_t r_{t+h}|\mathcal{F}_{t+h-1})) = \mathbb{E}(r_r \mathbb{E}(r_{t+h}|\mathcal{F}_{t+h-1})) = 0$. Since r_t assumes values in $(-1, 1)$, its unconditional variance is finite. Rocha and Cribari-Neto (2009) show that the error on the predictor scale has mean approximately equal to zero and variance approximately equal to $(\partial\mu_t/\partial\eta_t)^2[\mu_t(1-\mu_t)/(1+\phi)]$ and that such errors are approximately orthogonal.

The estimation of the parameters that index the βARMA model is typically performed by conditional maximum likelihood (Andersen, 1970). The conditional log-likelihood function (given the first K observations) is $\ell = \sum_{t=K+1}^{n} \log f(y_t|\mathcal{F}_{t-1})$, where $K = \max\{p, q\}$ and $f(y_t|\mathcal{F}_{t-1})$ is presented in Equation (1).

When the model contains moving average components, it is necessary to take into account the recursive structure of log-likelihood derivatives (Benjamin, Rigby, & Stasinopoulos, 1998). Using the predictor scale error, such derivatives are given by Rocha and Cribari-Neto (2017), as follows:

$$\frac{\partial \eta_t}{\partial \alpha} = 1 - \sum_{j=1}^{q} \theta_j \frac{\partial \eta_{t-j}}{\partial \alpha}, \quad \frac{\partial \eta_t}{\partial \beta} = \boldsymbol{x}_t' - \sum_{i=1}^{p} \varphi_i \boldsymbol{x}_{t-i}' - \sum_{j=1}^{q} \theta_j \frac{\partial \eta_{t-j}}{\partial \beta},$$

$$\frac{\partial \eta_t}{\partial \varphi_i} = g(y_{t-i}) - \boldsymbol{x}_{t-i}'\beta - \sum_{j=1}^{q} \theta_j \frac{\partial \eta_{t-j}}{\partial \varphi_i}, \quad i = 1, \ldots, p,$$

$$\frac{\partial \eta_t}{\partial \theta_l} = g(y_{t-l}) - \eta_{t-l} - \sum_{j=1}^{q} \theta_j \frac{\partial \eta_{t-j}}{\partial \theta_l}, \quad l = 1, \ldots, q.$$

Starting values for η_t can be obtained by setting $\eta_t = g(y_t)$ and the derivatives of η with respect to the model parameters equal to zero for $t = 1, \ldots, q$ (Benjamin et al., 1998).

Bayesian model selection for the βARMA model was developed by Casarin, Valle, and Leisen (2012), and bias-corrected maximum likelihood of the parameters that index the model was considered by Palm and Bayer (2018). An extension of the model that incorporates seasonal dynamics, the βSARMA model, was recently proposed by Bayer, Cintra, and Cribari-Neto (2018), and an extension of the model for compositional data, the DARMA model ("D" stands for Dirichlet), was developed by Zheng and Chen (2017). A dynamic model for doubly bounded random variables based on an alternative law—the Kumaraswamy law—was introduced by Bayer, Bayer, and Pumi (2017). In what follows, we shall focus on the standard, baseline βARMA model.

3 | STANDARD PORTMANTEAU TESTS

Portmanteau tests are commonly used in time series analysis to assess goodness-of-fit. The test statistics are based on residual autocorrelations, and the fitted model is taken as a good representation of the data when such autocorrelations are jointly negligible. In what follows, we shall briefly present some well-known portmanteau tests adapted for the βARMA model. Let w_t be a stationary time series of interest, and let $\rho_k = \text{cor}(w_t, w_{t+k})$ be the kth-order autocorrelation. For a user-selected integer $m > 0$, we are interested in testing

$$\begin{aligned} \mathcal{H}_0 &: \rho_1 = \rho_2 = \cdots = \rho_m = 0 \\ \mathcal{H}_1 &: \text{at least one } \rho_i \neq 0. \end{aligned} \tag{3}$$

That is, the interest lies in testing that the first m autocorrelations are jointly equal to zero.

Oftentimes, w_t is the tth residual from a fitted time series model. In that case, the null hypothesis under test is that the first m autocorrelations of the residuals from a fitted time series (ARMA or βARMA) are jointly equal to zero and its rejection is evidence of model misspecification. Let $\hat{r}_1, \ldots, \hat{r}_n$ denote the fitted model's residuals. The kth residual (sample) autocorrelation is $\hat{\rho}_k = \sum_{t=k+1}^{n} \hat{r}_t \hat{r}_{t-k} / \sum_{t=1}^{n} \hat{r}_t^2$, $k = 1, \ldots, m$. When the model is correctly specified, the residuals are expected to be nearly uncorrelated.

In Gaussian ARMA models, under the null hypothesis, the time series is a sequence of independent random variables, whereas in the βARMA model, such a series is comprised of noncorrelated variables.

Different residuals can be computed in the context of βARMA models. For instance, one can define residuals based on one the following discrepancies: $g(y_t) - g(\mu_t)$ and $y_t - \mu_t$. Recall that if, for example, w_t follows an ARMA process, then aw_t follows an ARMA process of the same order $\forall a \neq 0$. It is important to notice, however, that transformations of ARMA processes are generally not ARMA processes, and in the specific cases where they are, the order of the process usually changes (Linka, 1988). In the present context, we notice that, when $y_t \sim \beta\text{ARMA}(p, q)$ without covariates, conditionally on $\mathcal{F}_{t-1}$, the sequence $\{(g(y_t), r_t)\}_t$ satisfies the ARMA(p, q) difference equations. Working with residuals based on $y_t - \mu_t$ may be potentially problematic since $\{(y_t, r_t)\}_t$ may not satisfy the difference equations of an ARMA model. However, the asymptotic results we shall state later (Theorems 1 and 2) may still hold as the following simple, yet useful case exemplifies. Suppose that $y_t \sim \beta\text{ARMA}(p, q)$, consider the logit link function, that is, $g(x) = \log(x) - \log(1 - x)$, and consider the residual

$$\hat{r}_t = a_t(n)(y_t - \hat{\mu}_t) \tag{4}$$

for a sequence of random variables $\{a_t(n)\}_n$ such that, conditionally on $\mathcal{F}_{t-1}$, $a_t(n) \xrightarrow{p} a_t \in \mathbb{R}$ as n tends to infinity, for all t. By expanding $\log(1-x)$ into its power series around 0, we can write

$$g(y_t) - g(\mu_t) = y_t - \mu_t + \log\left(\frac{y_t}{\mu_t}\right) + \sum_{k=2}^{\infty} \frac{y_t^k - \mu_t^k}{k!}.$$

Notice that $0 < \mathrm{var}(y_t) \leq 1/[4(\phi+1)] < 1/4$, so that, if ϕ is large, the variance of y_t is very small and y_t will be close to μ_t with high probability. As a result, $\log(y_t/\mu_t) \approx 0$ and the summation, which involves powers of numbers in $(0, 1)$, can also be expected to be negligible. We thus expect that $y_t - \mu_t \approx g(y_t) - g(\mu_t)$. If, in addition (as often happens in applications), the sequence a_t is nearly constant, we expect that the behavior of $a_t(y_t - \mu_t)$ should not be far from a constant times $g(y_t) - g(\hat{\mu}_t)$. As a consequence, portmanteau tests based on $g(y_t) - g(\hat{\mu}_t)$ and on (4) should behave similarly.

We shall now present some portmanteau test statistics that were proposed in the context of Gaussian time series and that can be used with time series that follows the beta law. Box and Pierce (1970) introduced a test statistic that, under the null hypothesis, is approximately χ^2 distributed in large samples. The associated test, however, was shown to have poor small sample performance. A variant of the Box–Pierce test statistic was considered by Ljung and Box (1978), as follows:

$$Q_{\mathrm{LB}} = n(n+2)\sum_{k=1}^{m} \frac{\hat{\rho}_k^2}{n-k}.$$

Monti (1994) proposed replacing the sample autocorrelations in the above test statistic by sample partial autocorrelations. The limiting null distribution of both test statistics is χ^2_{m-p-q}.

A nonparametric portmanteau test statistic based on rank autocorrelations was introduced by Dufour and Roy (1986). Let R_t be the rank of $\hat{r}_t$. The kth residual rank autocorrelation is $\tilde{\rho}_k = \sum_{t=1}^{n-k}(R_t - \bar{R})(R_{t+k} - \bar{R}) / \sum_{t=1}^{n}(R_t - \bar{R})^2$, $1 \leq k \leq n-1$, where $\bar{R} = n^{-1}\sum_{t=1}^{n} R_t = (n+1)/2$ and $\sum_{t=1}^{n}(R - \bar{R})^2 = n(n^2-1)/12$ if all ranks are distinct. Since $R_1, \ldots, R_n$ are interchangeable, when $\hat{r}_1, \ldots, \hat{r}_n$ are interchangeable and continuous, the mean of $\tilde{\rho}_k$ is $\mu_k = -(n-k)/[n(n-1)]$ for $1 \leq k \leq n-1$. The authors also provide an expression for $\tilde{\sigma}_k^2$, the variance of $\tilde{\rho}_k$. The portmanteau test statistic is $Q_{\mathrm{DR}} = (\tilde{\boldsymbol{\rho}} - \boldsymbol{\mu})' D_2^{-1} (\tilde{\boldsymbol{\rho}} - \boldsymbol{\mu})$, where $\tilde{\boldsymbol{\rho}} = (\tilde{\rho}_1, \ldots, \tilde{\rho}_m)'$, $\mu = (\mu_1, \ldots, \mu_m)'$, and $D_2 = \mathrm{diag}\,\{\tilde{\sigma}_1^2, \ldots, \tilde{\sigma}_m^2\}$. Under $\mathcal{H}_0$, Q_{DR} is asymptotically distributed as χ^2_{m-p-q}.

Kwan and Sim (1996a) considered the situation in which a portmanteau test is applied to an observed time series, and not to residuals from a fitted model. They transformed the autocorrelations in order to reduce the dispersion bias of Q_{LB}. Fisher (1921) proposed transforming $\hat{\rho}_k$ as $z_{1k} = 0.5\log((1+\hat{\rho}_k)/(1-\hat{\rho}_k))$, and Hotelling (1953) introduced the following two transformations: $z_{2k} = z_{1k} - (3z_{1k} + \hat{\rho}_k)/[4(n-k)]$ and $z_{3k} = z_{2k} - 23z_{1k} + 33\hat{\rho}_k - 5\hat{\rho}_k^3/96(n-k)^2$, where z_{ik}, for $i = 1, 2, 3, 4$, is normally distributed with $\mathbb{E}(z_{ik}) \approx 0$, $\mathrm{var}(z_{1k}) \approx (n-k-3)^{-1}$, $\mathrm{var}(z_{2k}) \approx (n-k-1)^{-1}$, and $\mathrm{var}(z_{3k}) \approx (n-k-1)^{-1}$. Using these approximations, Kwan and Sim (1996a) proposed three modified portmanteau test statistics: $Q_{\mathrm{KW}i} = \sum_{k=1}^{m}(n-k-\tau_i)z_{ik}^2$, $i = 1, 2, 3$, where $\tau_1 = 3$ and $\tau_2 = \tau_3 = 1$. Under $\mathcal{H}_0$, they are asymptotically distributed as χ^2_m. The number of degrees of freedom is m because the test is, as noted above, applied to an observed time series.

Kwan and Sim (1996b) introduced a fourth test statistic based on a variance-stabilizing transformation proposed by Jenkins (1954): $z_{4k} = \sin^{-1}(\hat{\rho}_k)$. Here, $\mathbb{E}(z_{4k}) \approx 0$, and $\mathrm{var}(z_{4k}) \approx (n-k)^{-2}(n-k-1)$. The test statistic is $Q_{\mathrm{KW4}} = \sum_{k=1}^{m}[(n-k)^2/(n-k-1)]z_{4k}^2$. Under $\mathcal{H}_0$, it is asymptotically distributed as χ^2_m.

When the sample size is large relative to m, the means of Q_{KW1}, Q_{KW2}, and Q_{KW3} are approximately equal to $m - m(m+4)/n$, and the mean of Q_{KW4} is approximately equal to $m - m(m+1)/n$. Kwan and Sim (1996a) noticed that these results suggest that, for fixed n, the means of the four test statistics are smaller than m. (Such results were obtained for Gaussian processes. In our simulations, we computed the means of the four test statistics and noticed that the approximations also hold for βARMA processes.) The authors then proposed to modify the test critical values using

$$\mathbb{E}(Q_{\mathrm{KW}i}) = \sum_{k=1}^{m}(n-k-\tau_i)\left\{\mathbb{E}\left(\hat{\rho}_k^2\right) + \frac{2}{3}\mathbb{E}\left(\hat{\rho}_k^4\right)\right\}, \quad i = 1, 2, 3,$$

$$\mathbb{E}(Q_{\mathrm{KW4}}) = \sum_{k=1}^{m}\frac{(n-k)^2}{(n-k-1)}\left\{\mathbb{E}\left(\hat{\rho}_k^2\right) + \frac{1}{3}\mathbb{E}\left(\hat{\rho}_k^4\right)\right\}.$$

Expressions for $\mathbb{E}(\hat{\rho}_k^2)$ and $\mathbb{E}(\hat{\rho}_k^4)$ can be found in Davies, Triggs, and Newbold (1977) and in Ljung and Box (1978). The tests can then be performed as follows when applied to an observed time series: Reject the null hypothesis at the $\gamma \times 100\%$

significance level ($0 < \gamma < 1$) if $Q_{\text{KW}i} \geq \chi^2_{1-\gamma,\mathbb{E}(Q_{\text{KW}i})}$, $i = 1, \ldots, 4$. When the test statistics are computed from ARMA residuals, the null hypothesis is rejected if $Q_{\text{KW}i} \geq \chi^2_{1-\gamma,\mathbb{E}(Q_{\text{KW}i})-p-q}$, $i = 1, 2, 3, 4$.

Peña and Rodriguez (2002) proposed a different portmanteau test statistic that is based on the determinant of the residual autocorrelation matrix: $\hat{D}_m = n(1 - |\hat{R}_m|^{1/m})$, where $\hat{R}_m$ is the $(m+1) \times (m+1)$ sample autocorrelation matrix. The determinant of $\hat{R}_m$, $|\hat{R}_m|$, is the estimated generalized variance of the standardized residuals. The authors proposed approximating the asymptotic null distribution of $\hat{D}_m$ by the gamma distribution. The approximation, however, can be quite poor. A modified test statistic was also proposed by the authors: $D_m = n(1 - |\ddot{R}_m|^{1/m})$, where $\ddot{R}_m$ is obtained by replacing $\hat{\rho}_k^2$ with $\ddot{\rho}_k^2 = (n+2)(n-k)^{-1}\hat{\rho}_k^2$. McLeod and Jimenez (1984) noted a shortcoming of D_m: $\ddot{R}_m$ is not always positive definite.

Lin and McLeod (2006) recommended performing the Peña–Rodriguez test using bootstrap resampling. The numerical evidence in their paper shows that the test performs considerably better when one does so.

4 | NEW PORTMANTEAU TESTS

In this section, we introduce two new test portmanteau statistics that are based on residual partial autocorrelations. We then prove that their limiting null distribution is χ^2_{m-p-q} in the class of βARMA models. When the statistics are computed using observed time series, their limiting distribution under the null hypothesis of no serial correlation is χ^2_m. A novel aspect of our proofs is that they also hold for other test statistics that were proposed in the Gaussian time series literature, which are asymptotically equivalent to our test statistics under the null hypothesis.

4.1 | Two new test statistics

We shall now propose two new portmanteau test statistics, namely, Q_1 and Q_4. The new test statistics are based on residual partial autocorrelations. The motivation for using partial autocorrelations in portmanteau test statistics stems from the fact that when the order of moving average dynamics is underestimated, the sum of squared partial autocorrelations is likely to be larger than that of squared autocorrelations (see, e.g., Monti, 1994, pp. 778–779).

As before, let $\hat{\pi}_k$ denote the kth residual partial autocorrelation. We shall explore the use of the aforementioned variance-stabilizing transformations, but now applied to partial autocorrelations. We shall then use the transformed partial autocorrelations to construct portmanteau test statistics. At the outset, we consider the transformation introduced by Fisher (1921), as follows:

$$\hat{z}_{1k} = \frac{1}{2}\log\left(\frac{1+\hat{\pi}_k}{1-\hat{\pi}_k}\right), \quad k = 1, \ldots, m.$$

The corresponding modified test statistic can be written as

$$Q_1 = \sum_{k=1}^{m}(n-k-3)\hat{z}_{1k}^2. \tag{5}$$

The second test statistic we propose makes use of the transformation introduced by Jenkins (1954), namely,

$$\hat{z}_{4k} = \sin^{-1}(\hat{\pi}_k).$$

Using it, we arrive at the following portmanteau test statistic:

$$Q_4 = \sum_{k=1}^{m}\frac{(n-k)^2}{n-k-1}\hat{z}_{4k}^2. \tag{6}$$

We shall prove that, under appropriate conditions and under the null hypothesis, when the test is applied to an observed time series, Q_1 and Q_4 are asymptotically distributed as χ^2_m; when the test statistics are computed using βARMA residuals, they are, under the null hypothesis, asymptotically distributed as χ^2_{m-p-q}. Such results are proved in Section 4.2. We suggest that the test critical values be corrected using the approach outlined in the previous section.

The finite-sample performances of the two tests introduced above can be improved with the aid of bootstrap resampling. A similar approach can be applied to other portmanteau tests. In what follows, we shall use the bootstrap method to improve the finite-sample performances of the following tests: Q_{LB}, Q_{M}, Q_{DR}, Q_{KW1}, Q_{KW2}, Q_{KW3}, Q_{KW4}, Q_1, and Q_4. Bootstrap inference is performed as described in Lin and McLeod (2006).

4.2 | Asymptotic null distribution of the new portmanteau test statistics

We shall now derive the asymptotic distribution of the statistics Q_1 and Q_4 given in (5) and (6), respectively, under the null hypothesis (3) of no correlation up to a predetermined lag $m > 0$. We separate the discussion into two cases: The test is applied to an observed time series, and the test is applied to the residuals from a fitted βARMA model. As we have mentioned heuristically before, the asymptotic null distributions are different under these two cases. More specifically, when the time series under evaluation comes from a fitted model, it is obtained from a model with parameters replaced by estimates that impact the number of degrees of freedom of the test statistic asymptotic null distribution (for further details, see Box & Pierce, 1970; Ljung & Box, 1978; Ljung, 1986, as well as the references therein).

4.2.1 | Observed time series

We start by analyzing the case where the time series under scrutiny is an observed time series; that is, it is not a set of residuals from a time series model fit. Under the null hypothesis, the series is comprised of uncorrelated and identically distributed random variables with finite second moment. As before, for $k > 0$, let $\hat{\rho}_k$ and $\hat{\pi}_k$ denote the kth-order sample autocorrelation and the kth sample partial autocorrelation, respectively. Note that both $\hat{\rho}_k$ and $\hat{\pi}_k$ depend on n.

Theorem 1. *Under the null hypothesis, for* $i = 1, 4$*, the statistics* Q_i *and* $Q_{\text{KW}i}$ *are asymptotically equivalent, and*

$$Q_i \xrightarrow{d} \chi^2_m,$$

as n tends to infinity.

The proof of the above theorem and the proofs of the results that follow are presented in the Appendix.

4.2.2 | βARMA residuals

In order to obtain the asymptotic null distribution of the proposed test statistics, we need to rely on the asymptotic theory for the conditional maximum likelihood estimator (CMLE) in the framework of βARMA models. The asymptotic theory is mentioned in Rocha and Cribari-Neto (2009), and more details are given in Rocha and Cribari-Neto (2017). The idea behind the asymptotic theory of the CMLE in the context of βARMA models is to write η_t as a linear combination of the regressors x_t and past values of the process itself (in autoregressive fashion), allowing for possible ancillary terms, in the spirit of Benjamin et al. (2003; Equation (3)) and Fokianos and Kedem (2004; Equation (5) in particular).

In what follows, we summarize the conditions required for the consistency and asymptotic normality of the CMLE in βARMA models. We start with conditions regarding the model's systematic component.

- A1. The roots of the autoregressive polynomial $\Phi(z) = 1 - \varphi_1 z - \cdots - \varphi_p z^p$ lie outside the unit circle.
- A2. In the parametric space, there exists an open neighborhood U around the true parameters $\boldsymbol{\theta}$ where the roots of the moving average polynomial $\Theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$ lie outside the unit circle.
- A3. The polynomials $\Theta(\cdot)$ and $\Phi(\cdot)$ have no common roots.

Assumptions A1–A3 are standard in the context of usual ARMA models, and they provide sufficient smoothness for the CMLE's asymptotic theory to hold. Now, let $\boldsymbol{\kappa} = (\phi, \alpha, \beta_1, \dots, \beta_s, \varphi_1, \dots, \varphi_p, \theta_1, \dots, \theta_q)'$ and $\boldsymbol{d}_t = (\boldsymbol{x}'_t, \dots, \boldsymbol{x}'_1, \dots, y_t, \dots, y_1)'$, where x'_t is an s-dimensional set of nonrandom covariates. In addition to Conditions A1–A3, we make the following assumptions.

- P1. The inverse link function g^{-1} is of class C^2 and satisfies $|\partial g^{-1}(x)/\partial x| \neq 0$, for all $x \in \mathbb{R}$.
- P2. The parametric space Ω is an open set in $\mathbb{R}^{s+q+p+2}$, and the true parameter $\boldsymbol{\kappa}_0$ lies in Ω.
- P3. For each t, $\boldsymbol{d}_t$ almost surely belongs to a compact set Υ, and there exists $n_0 \in \mathbb{N}$ such that, for all $n > n_0$, $\sum_{t=1}^{n} \boldsymbol{d}_t \boldsymbol{d}'_t$ is positive definite with probability 1. Additionally, $g^{-1}(\eta_t)$ is almost surely well defined for all $\boldsymbol{d}_t \in \Upsilon$ and $\boldsymbol{\kappa} \in \Omega$.

P4. There exists a probability measure λ such that $\int \boldsymbol{z}\boldsymbol{z}'\lambda(d\boldsymbol{z})$ is positive definite and such that the weak convergence

$$\frac{1}{n}\sum_{t=1}^{n} I(\boldsymbol{d}_{t-1} \in A) \underset{n\to\infty}{\longrightarrow} \lambda(A)$$

holds for all λ-continuity sets A under (2) with $\boldsymbol{\kappa} = \boldsymbol{\kappa}_0$.

Under A1–A3 and P1–P4, the CMLE is consistent and asymptotically normal with the variance–covariance matrix equal to the inverse of the information matrix. For further details and for a discussion of the conditions above, we refer readers to the works of Fahrmeir and Kaufmann (1985); Shao (1992); Fokianos and Kedem (2004); and Pumi, Valk, Bisognin, Bayer, and Prass (2019) as well as the references therein.

To set the notation, let $\hat{r}_1, \dots, \hat{r}_n$ be the residuals obtained from a correctly specified βARMA(p, q) model, computed as $\hat{r}_t = g(y_t) - g(\hat{\mu}_t)$, where $\hat{\mu}_t$ is obtained by evaluating (2) at the maximum likelihood estimates.

Theorem 2. *Let Q_i, $i = 1, 4$, be computed using residuals obtained from a correctly specified βARMA(p, q) model that satisfies Assumptions A1–A3 and P1–P4. Under the null hypothesis,*

$$Q_i \xrightarrow{d} \chi^2_{m-p-q},$$

as n tends to infinity.

Using the proof of the above result, it is possible to establish the limiting null distribution of Q_{LB} and Q_{M} when such test statistics are computed using βARMA residuals, as indicated in the next result.

Corollary 1. *Let Q_{LB} and Q_{M} be computed using residuals from a correctly specified βARMA(p, q) model that satisfies Assumptions A1–A3 and P1–P4. Under the null hypothesis,*

$$Q_{\mathrm{LB}} \xrightarrow{d} \chi^2_{m-p-q} \quad \textit{and} \quad Q_{\mathrm{M}} \xrightarrow{d} \chi^2_{m-p-q},$$

as n tends to infinity.

We close this section by noting that one can force b coefficients to equal zero when fitting a βARMA model ($b < p + q$), obtain the residuals, and perform portmanteau testing inferences. In that case, the asymptotic null distribution of Q_1 and Q_4 is $\chi^2_{m-p-q+b}$. The proof is essentially the same as the one presented in the Appendix, in view of McLeod (1978).

5 | NUMERICAL EVIDENCE

Several simulation experiments were carried out to investigate the finite-sample performances of the different portmanteau tests in the class of βARMA models. All simulations were performed using the R statistical computing environment (R Core Team, 2019). We use the standardized ordinary residual defined by Ferrari and Cribari-Neto (2004), that is, $\hat{r}_t = (y_t - \hat{\mu}_t)/\sqrt{\widehat{\mathrm{var}}(y_t)}$, where $\widehat{\mathrm{var}}(y_t) = \hat{\mu}_t(1 - \hat{\mu}_t)/(1 + \hat{\phi})$. Here, $\hat{\mu}_t$ is obtained by evaluating μ_t at the maximum likelihood estimates, and $\hat{\phi}$ is the maximum likelihood estimate of ϕ. In the simulations that follow, we use $\phi = 120$ so that $\mathrm{var}(y_t) < 1/484 \approx 0.00207$. Under the conditions stated in Section 4.2, if we take $a_t(n) = \widehat{\mathrm{var}}(y_t)^{-\frac{1}{2}} \xrightarrow{p} \mathrm{var}(y_t)^{-\frac{1}{2}} = a_t$, for all t, as n tends to infinity; recall (4). It follows that Theorem 2 holds.

All tests are performed at 10%, 5%, and 1% significance levels. For brevity, however, we shall only present results for $\gamma = 5\%$. The sample sizes are $n = 50, 250, 500$; the values of m we used are $m = 5, 10, 15, 20, 25$; and the number of Monte Carlo replications is 5,000. Separate simulations were performed for each value of m. Log-likelihood maximization was performed using the Broyden–Fletcher–Goldfarb–Shanno quasi-Newton method with analytic first derivatives. Starting values for the parameters were selected as follows: (a) All moving average parameters were set equal to zero, and (b) the values for the autoregressive parameters were selected by regressing $g(y_t)$ on $g(y_{t-1}), \dots, g(y_{t-p})$ using ordinary least squares. Beta random number generation was performed based on the Mersenne Twister uniform random generator (Matsumoto & Nishimura, 1998). All simulations were carried out using the logit link function.

Our Monte Carlo simulations entailed considerable computational cost. They were run at the Centro Nacional de Supercomputação of Universidade Federal do Rio Grande do Sul. The hardware used was a cluster of computers with 64 blades of processing, 15.97 TFLOPS, and 174-TB RAM running the SUSE Linux Enterprise Server operating system. Our code made use of parallel computing, and our simulations ran on three nodes with 24 clusters. To achieve reproducibility, we used the doRNG R package in conjunction with `foreach` loops. By using parallel computing, we were able to reduce execution time by approximately 78%.

At the outset, we ran size simulations using asymptotic critical values. We considered the following data-generating mechanisms: βAR(1) (with $\varphi = 0.2, 0.5, 0.8$), βMA(1) (with $\theta = 0.2, 0.5, 0.8$), and βARMA(1, 1) (with $\varphi = 0.2$ and $\theta = 0.2, 0.5, 0.8$). The tests displayed considerable size distortions in some cases when the sample size was small ($n = 50$) but performed well with larger sample sizes ($n = 250$ or $n = 500$). We shall not present these results for brevity. Instead, we shall focus on size simulations of tests that employ bootstrap resampling. We note that small-sample-size distortions also take place when the tests are used with Gaussian ARMA models. Lin and McLeod (2006), for example, recommend using bootstrap resampling when performing the portmanteau test proposed by Peña and Rodriguez (2002) with Gaussian models when $n < 1,000$.

We shall now investigate the effectiveness of bootstrap resampling when coupled with portmanteau tests in the class of βARMA models. We shall consider the following bootstrap tests: Q_{LB}, Q_{M}, Q_{DR}, Q_{KW1}, Q_{KW2}, Q_{KW3}, Q_{KW4}, Q_1, and Q_4. Since Q_{KW1}, Q_{KW2}, and Q_{KW3} behave similarly, we shall only report results for Q_{KW1}. All results are based on 1,000 bootstrap samples and $n = 50$. We shall not report results for $n \in \{250, 500\}$ because, in large samples, the bootstrap tests behave similarly to the corresponding standard tests.

Table 1 contains the null rejection rates of the bootstrap portmanteau tests, including the bootstrap variant of the Peña–Rodriguez test (Lin & McLeod, 2006), for the βAR(1) model with $\varphi \in \{0.2, 0.5, 0.8\}$. Table 2 presents the bootstrap test null rejection rates obtained using the βMA(1) model with $\theta \in \{0.2, 0.5, 0.8\}$ for data generation. In both cases, all tests are now nearly size distortion free.

A second set of Monte Carlo simulations was carried out to evaluate the tests' nonnull behavior, that is, to evaluate the tests' powers. Data generation is now carried out under the alternative hypothesis, and the interest lies in examining the tests' ability to detect that the model specification is in error. The true data-generating process is βARMA(1, 1), and the fitted model is βAR(1). The sample sizes are $n \in \{50, 250\}$, and the values of m range from 3 to 25. Since Q_{KW1}, Q_{KW2}, Q_{KW3}, and Q_{KW4} behave similarly, we shall only report results on Q_{KW4}. The Q_1 and Q_4 tests proposed in this paper also

TABLE 1 Null rejection rates of the bootstrap portmanteau tests at the 5% nominal level obtained using the βAR(1) model with $\varphi = 0.2, 0.5, 0.8$ and $n = 50$

Test	$\varphi = 0.2$					$\varphi = 0.5$					$\varphi = 0.8$				
	m					m					m				
	5	10	15	20	25	5	10	15	20	25	5	10	15	20	25
$\hat{D}_m$	4.8	4.9	5.3	5.4	5.0	4.1	5.0	5.1	5.0	4.6	4.7	4.8	4.6	4.6	4.8
Q_{LB}	5.0	5.0	4.8	4.8	4.9	4.7	5.0	5.0	5.5	5.2	5.1	4.8	4.9	4.7	4.8
Q_{M}	5.0	4.9	5.2	5.2	5.1	4.6	5.2	5.0	5.0	5.5	5.1	5.2	4.7	5.1	4.8
Q_{DR}	5.3	4.8	5.0	4.6	4.8	5.1	5.1	5.1	4.8	5.0	5.0	4.9	4.7	5.0	5.0
Q_{KW1}	5.0	5.0	4.7	4.5	4.9	4.7	5.0	5.1	5.3	5.3	5.0	4.9	4.7	5.0	5.0
Q_{KW4}	5.0	5.0	4.7	4.6	4.8	4.7	5.0	5.1	5.3	5.3	5.0	4.9	4.7	5.0	5.0
Q_1	5.1	4.9	5.2	4.9	5.2	4.6	4.9	5.2	5.0	5.3	4.9	5.0	4.8	5.1	5.2
Q_4	5.1	4.9	5.2	5.0	5.3	4.6	4.9	5.2	5.0	5.2	4.9	5.0	4.9	5.2	5.2

TABLE 2 Null rejection rates of the bootstrap portmanteau tests at the 5% nominal level obtained using the βMA(1) model with $\theta = 0.2, 0.5, 0.8$ and $n = 50$

Test	$\theta = 0.2$					$\theta = 0.5$					$\theta = 0.8$				
	m					m					m				
	5	10	15	20	25	5	10	15	20	25	5	10	15	20	25
$\hat{D}_m$	4.8	5.2	5.4	5.3	5.2	4.5	4.9	4.7	5.0	4.9	4.8	5.2	5.1	5.0	5.0
Q_{LB}	5.1	5.0	5.0	5.0	4.8	5.4	4.8	5.1	5.0	5.2	5.3	5.1	5.3	5.6	5.6
Q_{M}	5.1	4.8	5.0	5.3	5.2	5.4	4.9	5.3	5.3	5.2	5.3	4.9	4.9	5.4	5.3
Q_{DR}	5.1	5.1	4.8	5.0	4.6	5.0	4.8	4.8	4.6	4.6	4.9	5.3	5.0	5.0	5.0
Q_{KW1}	5.3	5.0	5.2	5.2	5.0	5.4	4.9	4.9	5.1	5.0	5.4	5.1	5.2	5.4	5.6
Q_{KW4}	5.3	5.0	5.2	5.2	5.0	5.4	4.9	5.0	5.1	5.1	5.4	5.1	5.1	5.4	5.6
Q_1	5.1	4.8	5.1	5.3	5.0	5.4	4.8	4.9	5.2	5.2	5.2	4.7	4.9	5.2	5.2
Q_4	5.1	4.8	5.1	5.3	5.0	5.4	4.8	4.9	5.3	5.2	5.2	4.7	4.9	5.2	5.0

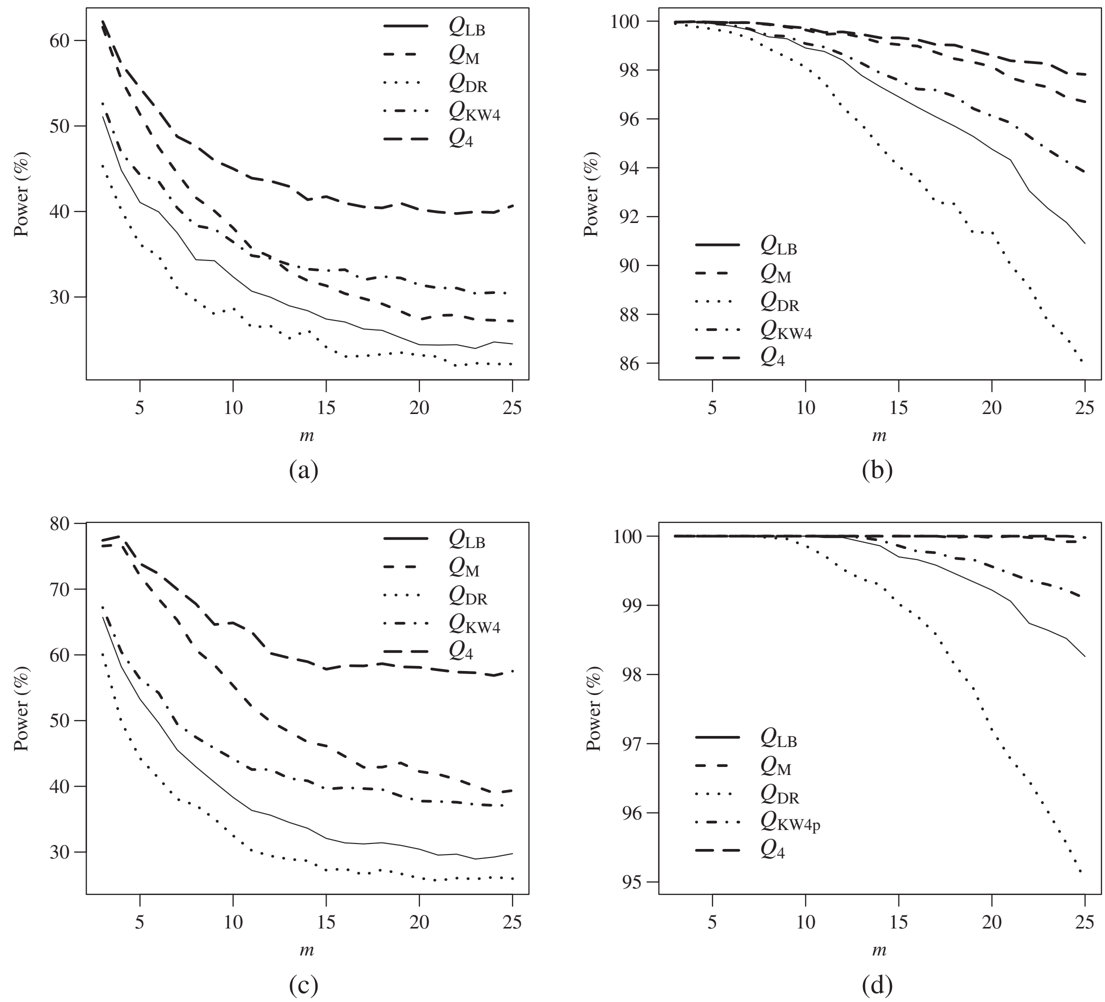

FIGURE 1 Powers of the Q_{LB}, Q_M, Q_{DR}, Q_{KW4}, and Q_4 tests at the 5% nominal level when the fitted model is βAR(1) and the true data-generating process is βARMA(1, 1) with $\varphi = 0.5$, $\theta = 0.5$ (top row) and $\varphi = 0.2$, $\theta = 0.8$ (bottom row). (a) $n = 50$. (b) $n = 250$. (c) $n = 50$. (d) $n = 250$

behave similarly, and for that reason, we shall only consider Q_4. Since some of the tests are liberal, we shall base all tests on exact (estimated from the size simulations) critical values rather than on asymptotic critical values. By doing so, we force all tests to have correct size.

The plots in the top row of Figure 1 display the empirical powers of Q_{LB}, Q_M, Q_{DR}, Q_{KW4}, and Q_4 for $\varphi = 0.5$ and $\theta = 0.5$. Notice that when $n = 50$ (left panel), Q_4, the test proposed in this paper, outperforms the remaining tests for all values of m. When the sample size is large ($n = 250$, right panel), the Q_4 test considerably outperforms all other tests for all values of m. The Q_{DR} test is the worst performer in both cases ($n = 50$ and $n = 250$). The plots in the bottom row of Figure 1 present results obtained using $\varphi = 0.2$ and $\theta = 0.8$ in the βARMA data-generating process. The conclusions are similar to those from the previous case with Q_4 outperforming the competition regardless of the sample size. A visual inspection of the graphics in the bottom row of Figure 1 reveals that the choice of m considerably impacts the powers of the tests. In particular, the tests' ability to detect model misspecification weakens as m grows. The same happens under Gaussian models (Kwan, Sim, & Wu, 2005).

How does the value of θ impact the tests' powers? In order to answer that question, we ran simulations using different values of the moving average parameter when generating the data. The value of m is fixed at 5, the value of the autoregressive parameter (φ) is fixed at 0.2, and two sample sizes are used: $n \in \{50, 250\}$. The tests' estimated powers are displayed in Figure 2 (left panel for $n = 50$ and right panel for $n = 250$). The Q_4 and Q_M tests are the clear winners when the sample size is small, especially when the value of θ is large; Q_{DR} is the test with the smallest power. When the sample size is large, Q_4 and Q_M remain the most powerful tests, but not by much, and Q_{KW4} becomes the worst performing test.

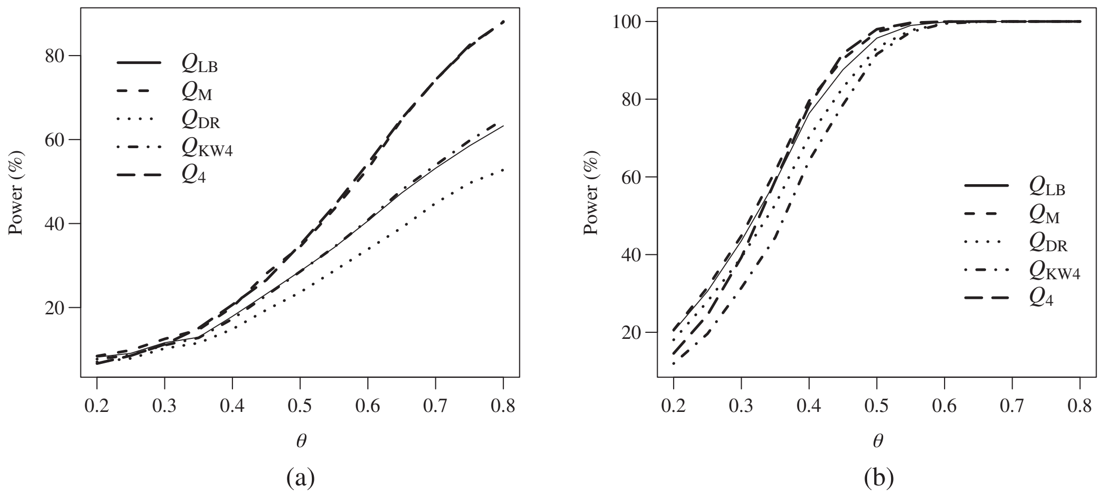

FIGURE 2 Powers of the Q_{LB}, Q_M, Q_{DR}, Q_{KW4}, and Q_4 tests at the 5% nominal level when the fitted model is βAR(1) and the true model is βARMA(1, 1) with $\varphi = 0.2$ and different values of θ. (a) $n = 50$. (b) $n = 250$

We now move to the situation where the true data-generating process is βARMA(1, 1) but the fitted model is βMA(1). In the previous case, the fitted model was incorrectly specified because it failed to take into account relevant moving average dynamics. In contrast, model misspecification now stems from failing to account for important autoregressive dynamics. For brevity, we shall only consider two scenarios. The tests' powers are displayed in plots in the top row of Figure 3, which shows the estimated powers of Q_{LB}, Q_M, Q_{DR}, Q_{KW4}, and Q_4 for $\varphi = 0.5$ and $\theta = 0.5$. The tests' ability to detect model misspecification increases with the sample size, as expected, and decreases with m. When the sample size is small (left panel), Q_{KW4} is the most powerful test, followed by Q_4 when m is large. When the sample size is large, Q_{KW4} and Q_4 are the winners, but not by much when m is small.

We shall now investigate whether the previous set of results is sensitive to the parameter values in the autoregressive and moving average dynamics. The plots in the middle row of Figure 3 present the tests' empirical powers for $n = 50$: (a) $\varphi = 0.5$ and $\theta = 0.8$ (left panel) and (b) $\varphi = 0.8$ and $\theta = 0.5$ (right panel). As in the previous case, Q_{KW4} is the most powerful test, with Q_4 being the runner-up when $m > 10$ in the left panel and when $m > 16$ in the right panel. Overall, these results are similar to those presented in the previous figure.

Figure 3 (bottom row, left panel) presents the empirical powers of the Q_{LB}, Q_M, Q_{DR}, Q_{KW4}, and Q_4 tests for $\varphi = 0.8$ and $\theta = 0.2$. The results obtained using $\varphi = 0.8$ and $\theta = 0.8$ are displayed in Figure 3 (bottom row, right panel). In both cases, $n = 50$ and Q_{KW4} is the winner. In the right panel, Q_4 is the runner-up, and in the left panel, Q_4 is the second-best performing test when m is large. Interestingly, in both cases, Q_M is the worst performer. Recall that the Q_M and Q_4 test statistics use residual partial autocorrelations. Even though we do not present results for $n = 250$, we note that when $\varphi = 0.8$, all tests displayed power of 100%.

Previously, we evaluated the tests' nonnull behavior for different values of the moving average parameter. We shall now evaluate how the tests' powers change with the value of the autoregressive parameter. Here, the data-generating process is βARMA(1, 1), the βARMA(0, 1) model is estimated, $\theta = 0.8$, and $m = 25$. The tests' powers for $n = 50, 250$ and different values of φ are presented in Figure 4. When $n = 50$ (left panel), Q_{KW4} and Q_4 are the most powerful tests for all values of φ, especially when φ is large, and Q_M is the worst performer. When $n = 250$ (right panel), Q_{KW4} and Q_4 are the winners for all values of φ, but not by much.

We shall now consider a different model specification error, namely, the true data-generating process is βARMA(2, 1), but the fitted model is βARMA(1, 1); that is, some existing autoregressive dynamics are not accounted for. The plots in the top row of Figure 5 display the empirical powers of Q_{LB}, Q_M, Q_{DR}, Q_{KW4}, and Q_4 for $\boldsymbol{\varphi} = (0.2, 0.5)'$ and $\theta = 0.2$. All the tests become considerably more powerful when n goes from 50 to 250. Again, the tests' powers decrease with m. In both cases, Q_{KW4} is the clear winner.

The plots in the bottom row of Figure 5 display the empirical powers of Q_{LB}, Q_M, Q_{DR}, Q_{KW4} and Q_4 for $\varphi = (0.2, 0.8)'$ and $\theta = 0.2$. The Q_{KW4} test again outperforms the competition. It is noteworthy that Q_M (which, like Q_4, is based on residual partial autocorrelations) is the worst performer. We also note that Q_4 is the second-best performer when the sample size is small and m is large.

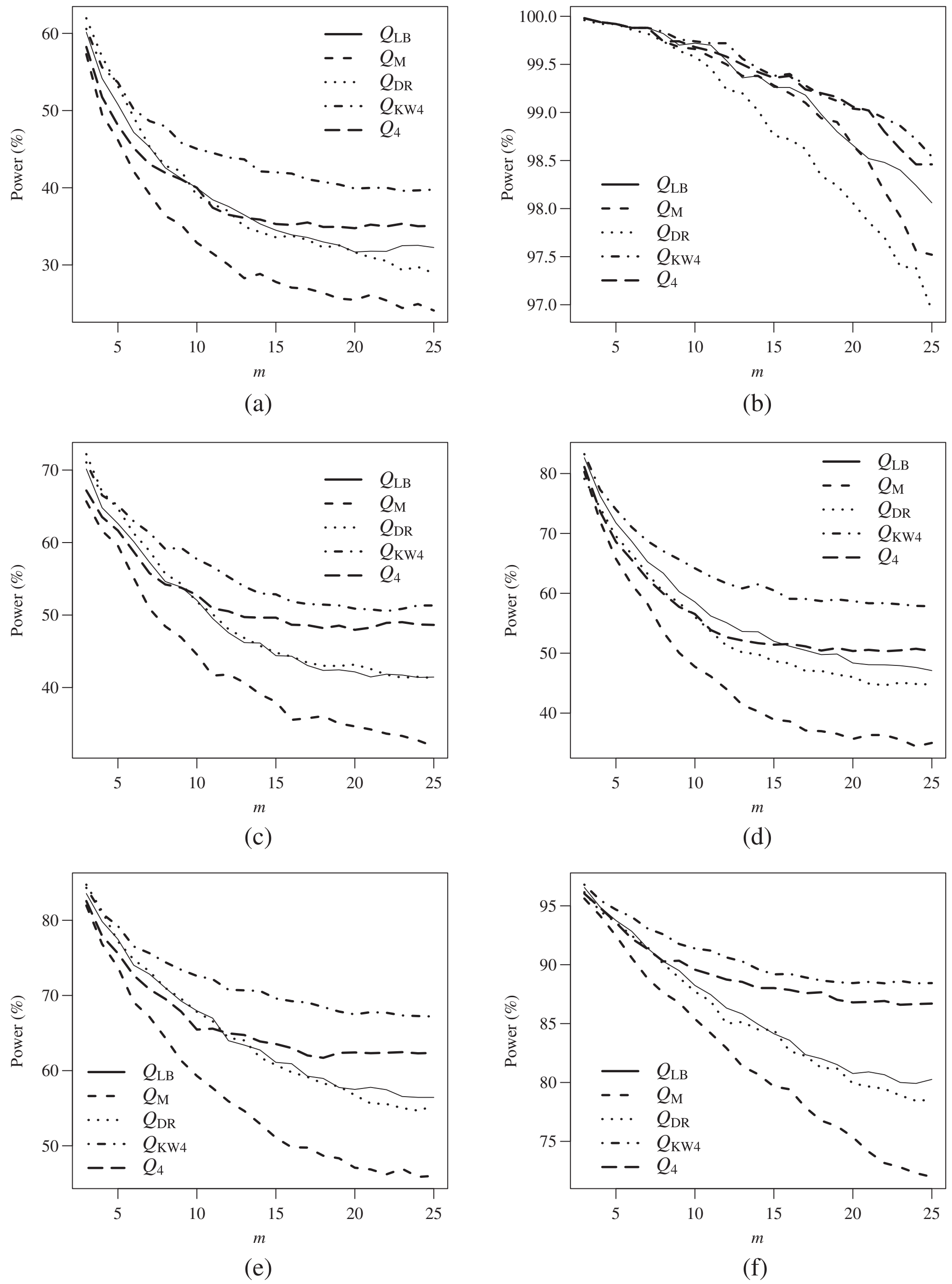

FIGURE 3 Powers of the Q_{LB}, Q_M, Q_{DR}, Q_{KW4}, and Q_4 tests at the 5% nominal level when the fitted model is βMA(1) and the true data-generating process is βARMA(1, 1) with $\varphi = 0.5$, $\theta = 0.5$ (top row); $\varphi = 0.5$, $\theta = 0.8$ (left panel, middle row); $\varphi = 0.8$, $\theta = 0.5$ (right panel, middle row); $\varphi = 0.8$, $\theta = 0.2$ (left panel, bottom row); and $\varphi = 0.8$, $\theta = 0.8$ (right panel, bottom row). (a) $n = 50$. (b) $n = 250$. (c) $n = 50$. (d) $n = 50$. (e) $n = 50$. (f) $n = 50$

Finally, we shall consider a data generation process of higher order. The data are generated using the βARMA(4, 4) dynamics, but the βARMA(2, 2) model is estimated. Here, $\boldsymbol{\varphi} = (0.15, 0.2, 0.25, 0.3)'$, and $\boldsymbol{\theta} = (0.07, 0.13, 0.21, 0.33)'$. The tests' powers for $n = 50$ were lower than in the previous simulations, with the best (worst) performing test being Q_4 (Q_{DR}) whose nonnull rejection rates are approximately equal to 25% (below 20%). In Figure 6, we present the tests' estimated powers for $n = 100, 250$. When $n = 100$, Q_{KW4} is the most powerful test for values of m up to 25, and Q_4 is the best performer for larger values of m. When $n = 250$, Q_{KW4} and Q_4 are clearly the most powerful tests for all values of m.

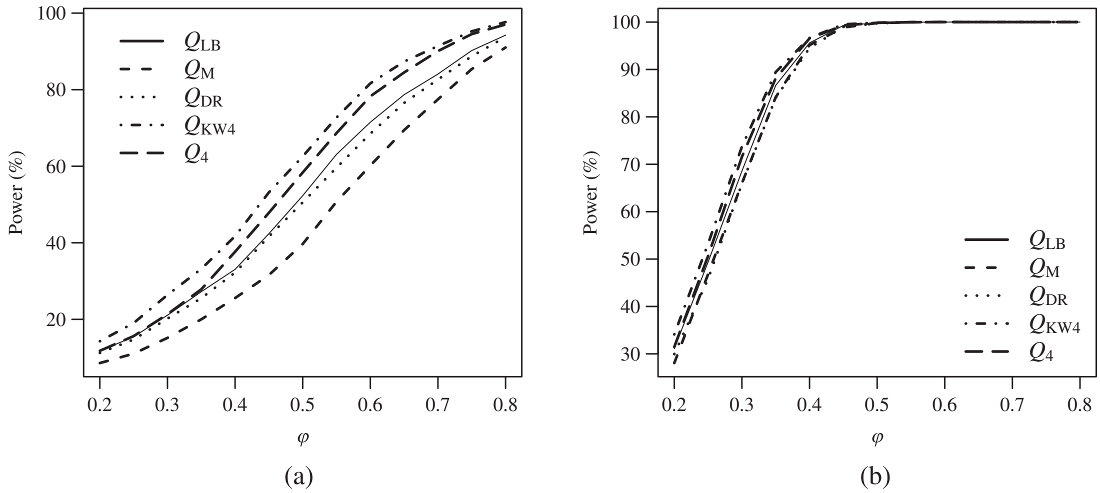

FIGURE 4 Powers of the Q_{LB}, Q_M, Q_{DR}, Q_{KW4}, and Q_4 tests at the 5% nominal level when the fitted model is βMA(1) and the true model is βARMA(1, 1) with $\theta = 0.8$ and different values of φ. (a) $n = 50$. (b) $n = 250$

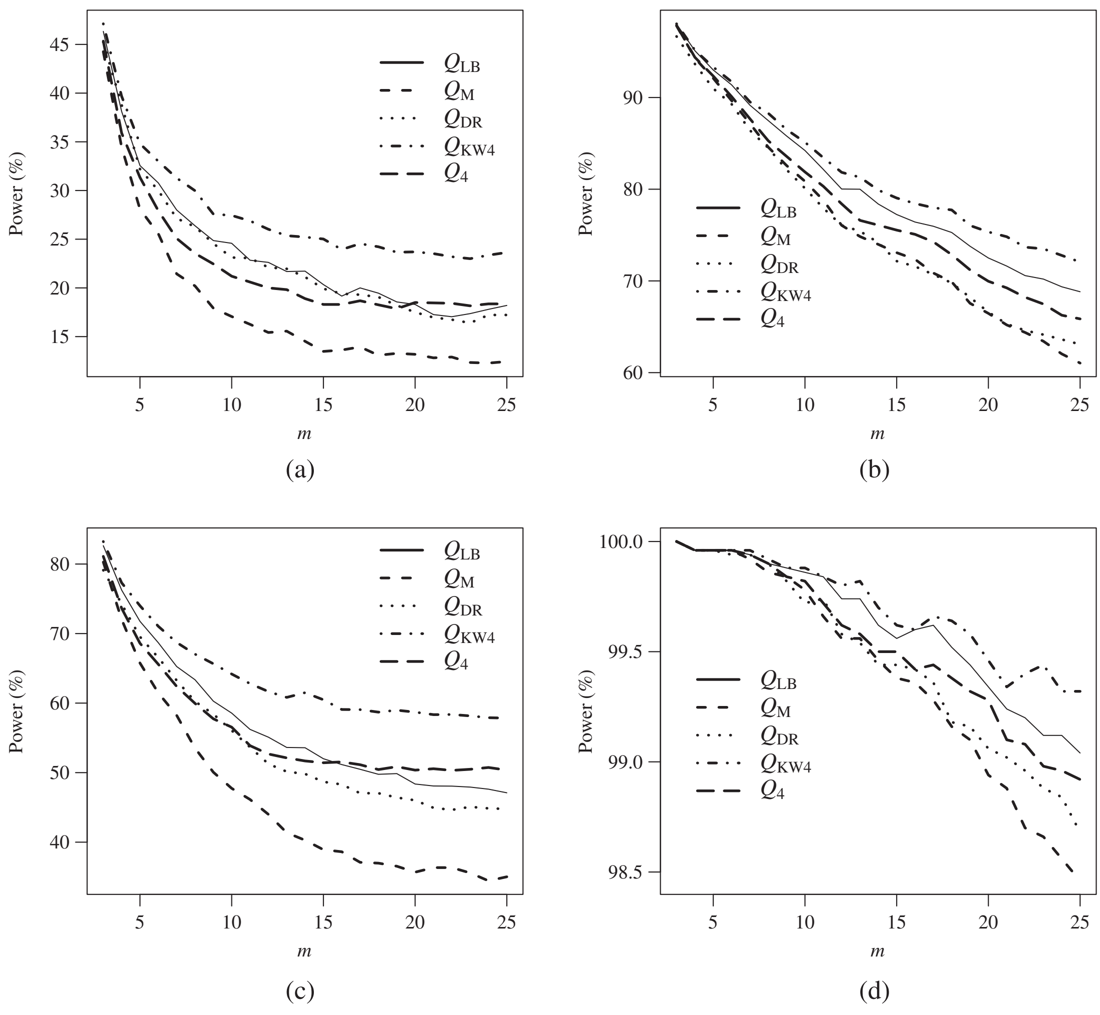

FIGURE 5 Powers of the Q_{LB}, Q_M, Q_{DR}, Q_{KW4}, and Q_4 tests when the fitted model is βARMA(1, 1) and the true model is βARMA(2, 1) with $\boldsymbol{\varphi} = (0.2, 0.5)'$, $\theta = 0.2$ (top row) and $\varphi = (0.2, 0.8)'$, $\theta = 0.2$ (bottom row). (a) $n = 50$. (b) $n = 250$. (c) $n = 50$. (d) $n = 250$

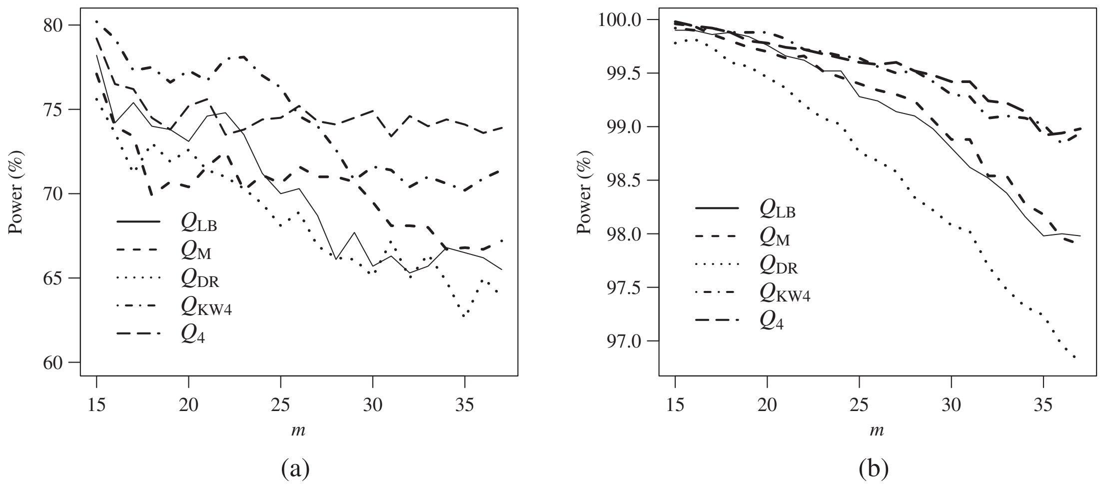

FIGURE 6 Powers of Q_{LB}, Q_{M}, Q_{DR}, Q_{KW4}, and Q_4 tests at the 5% nominal level when the fitted model is βARMA(2, 2) and the true model is βARMA(4, 4) with $\boldsymbol{\varphi} = (0.15, 0.2, 0.25, 0.3)'$ and $\boldsymbol{\theta} = (0.07, 0.13, 0.21, 0.33)'$. (a) $n = 100$. (b) $n = 250$

TABLE 3 Descriptive statistics of the monthly average rates of stocked energy in the South of Brazil

Min	Max	Median	Mean	Variance	Asymmetry	Excess kurtosis
0.2977	0.9862	0.7323	0.7069	0.0403	−0.3270	−1.1644

6 | EMPIRICAL HYDROLOGICAL APPLICATION

We shall now turn to the empirical application briefly described in Section 1. The variable of interest is the proportion of stocked hydroelectric energy (Operador Nacional do Sistema Elétrico, 2016) in South Brazil. The data are monthly averages from January 2001 to October 2016, thus covering 190 months ($n = 190$). The following six observations (November 2016 through April 2017) were used for evaluating forecasting accuracy. The computer code used in this case study is available at https://github.com/vscher/barma.

Table 3 contains some descriptive statistics. Notice the negative skewness and the negative excess kurtosis, and recall that the beta density easily accommodates such features.

Model selection was performed using the Akaike information criterion (AIC; Akaike, 1974). We considered all models with autoregressive and moving average dynamics up to the fourth order and logit link. The selected model was the βARMA(1, 1) model, whose AIC is equal to -307.9635. The maximum likelihood estimates of α, φ_1, θ_1, and ϕ are (standard errors in parentheses) 0.3452 (0.0787), 0.5235 (0.0412), 0.3588 (0.0502), and 11.7593 (1.1910), respectively. All parameters are significantly different from zero at the 1% significance level using the z test.

We shall now consider portmanteau testing inference; that is, we shall use the different portmanteau tests to assess whether the fitted model adequately represents the time series data. We computed the tests' p values for $m \in \{3, \ldots, 30\}$. The only p values smaller than .05 are those of the Q_{DR} test for small values of m (for values of m up to 5). Except for that, the p values of all tests for all values of m are greater than .05. We take as evidence that the βARMA(1, 1) model is correctly specified.

Next, we shall perform portmanteau testing inference using a different model. Since the βARMA(1, 1) model seems to adequately represent the data dynamics, the tests are expected to detect model misspecification when applied to a different model. At the outset, we fitted the βAR(1) model. Table 4 displays the tests' p values for $m \in \{5, 10, 15, 20, 25, 30\}$. The Q_4 test is the only test with all p values below .05; that is, it is the only test that yields rejection of the null hypothesis of

TABLE 4 Q_{LB}, Q_{M}, Q_{DR}, Q_{KW1}, Q_{KW4}, and Q_4 p-values computed from the fitted βAR(1) model

	$m = 5$	$m = 10$	$m = 15$	$m = 20$	$m = 25$	$m = 30$
Q_{LB}	0.006	0.006	0.026	0.058	0.128	0.182
Q_{M}	< 0.001	0.002	0.008	0.042	0.054	0.065
Q_{DR}	< 0.001	0.002	0.011	0.022	0.056	0.078
Q_{KW1}	0.005	0.005	0.021	0.043	0.087	0.120
Q_{KW4}	0.005	0.006	0.021	0.044	0.090	0.125
Q_4	< 0.001	0.002	0.006	0.029	0.038	0.045

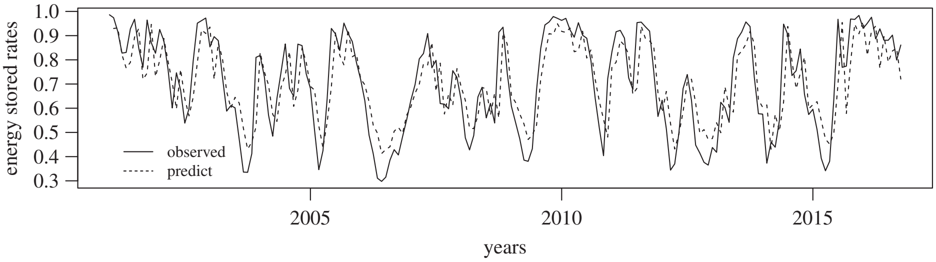

FIGURE 7 Energy stocked rates (solid lines) and forecasted values (dashed lines) computed from the fitted βARMA(1, 1) model

TABLE 5 Mean absolute forecasting errors from the βARMA(1, 1), ARMA(1, 1), AR(2), and KARMA(1, 1) models and from the Holt exponential smoothing algorithm

	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$
βARMA(1, 1)	0.1244	0.1444	0.1364	0.1484	0.1694	0.1839
ARMA(1, 1)	0.1518	0.1828	0.1820	0.1982	0.2211	0.2364
AR(2)	0.1345	0.1690	0.1680	0.1830	0.2050	0.2198
KARMA(1, 1)	0.1474	0.1788	0.1783	0.1950	0.2189	0.2352
Holt	0.1554	0.2110	0.2303	0.2629	0.2996	0.3264

correct model specification regardless of the value of m. When the βMA(1) model is fitted, all tests yield rejection of $\mathcal{H}_0$ at the 5% significance level for all values of m.

The final step in our empirical analysis involves forecasting. Indeed, stocked energy forecasting is quite important for all institutions responsible for energy distribution. We produced forecasts using three different time series models, namely, the βARMA(1, 1), Gaussian ARMA(1, 1), and Gaussian AR(2) models. The latter two models were selected based on the AIC using the `auto.arima` function of the forecast package (Hyndman & Khandakar, 2008) of the R statistical computing environment (R Core Team, 2019). We also used the KARMA(1, 1) model (Bayer et al., 2017); again, model selection was performed using the AIC. Finally, we considered the Holt exponential smoothing algorithm, as implemented in the `holt` function of the R forecast package. The observed time series and the predicted values from the fitted βARMA(1, 1) model are presented in Figure 7. It is noteworthy that the βARMA(1, 1) model is able to satisfactorily capture the data dynamics.

We now move from in-sample to out-of-sample forecasting. We consider a horizon of 6 months; that is, we wish to forecast the time series' next six values. Forecasting accuracy is measured using the mean absolute error (MAE), that is, the mean value of the absolute differences between observed and predicted values. The results are presented in Table 5 for $h = 1, \ldots, 6$, with h denoting the forecasting horizon. We note that the βARMA(1, 1) model yields forecasts that are more accurate than those obtained from the competing models and the smoothing algorithm for all values of h. For instance, when forecasting the next three observations ($h = 3$), the βARMA(1, 1) MAE is equal to 0.1364, which is considerably smaller than the MAEs of the three competing models (0.1820, 0.1680, and 0.1783) and of the exponential smoothing algorithm (0.2302).

7 | CONCLUDING REMARKS

The βARMA model is particularly useful for modeling and forecasting time series data that assume values in the standard unit interval. It is thus useful for modeling several hydrological time series. The model naturally accommodates distributional asymmetry and nonconstant variance. It will always yield fitted values and out-of-sample forecasts that are positive and smaller than 1. Additionally, no data transformation is needed prior to the analysis. The fitted model must be validated before it is used for forecasting. This is where our interest lies. Can the standard portmanteau tests be used with βARMA models? If so, how do they behave in finite samples? What is the impact of the choice of the truncation lag (m) on the tests' null and nonnull behaviors? We reviewed several portmanteau tests that are available in the literature and proposed two new tests. The test statistics we propose use residual partial autocorrelations instead of residual autocorrelations. We derived the asymptotic null distribution of the two new test statistics; more specifically, we proved that, under the null hypothesis, they are asymptotically distributed as χ^2_{m-p-q}. Our proof implies that some other well-known test statistics (Q_{BP}, Q_{LM}, and Q_{M}) are also asymptotically χ^2_{m-p-q} distributed under the null hypothesis when computed from βARMA residuals.

We presented Monte Carlo simulation results on the finite-sample behavior of the different portmanteau tests in the class of βARMA models. The tests' size distortions were small when bootstrap resampling was used, especially when m is not very small.

The most interesting evidence from our numerical evaluations, however, relates to the tests' powers. First, the choice of m impacts such powers: They typically decrease with m. Such an impact is well documented in the Gaussian literature (see, e.g., Kwan et al., 2005). Second, a portmanteau test that proved to be robust under Gaussian data did not perform well when used with βARMA models. Third, overall, the tests we proposed figure among the most powerful ones (the two new tests displayed similar nonnull behaviors; hence, we only presented results for one of them). In particular, they were the best performers under pure autoregressive dynamics. It is noteworthy that whenever our tests were not the most powerful ones, they were the next best performers as long as the value of m is not small. Fourth, overall, the tests we proposed proved to be more powerful than an existing portmanteau test that also uses residual partial autocorrelations. Of course, the evidence we report is restricted to the models considered in our numerical evaluation.

We also presented and discussed an empirical hydrological application. Our focus was on modeling and forecasting the proportion of stocked hydroelectric energy in the Southern Region of Brazil. Such an empirical application showed the usefulness of portmanteau testing inference for model validation and the usefulness of the class of βARMA time series models. It is noteworthy that the βARMA model used in the application yielded out-of-sample forecasts that were more accurate than those obtained using three alternative time series models and an exponential smoothing algorithm.

In future work, we shall consider the use of the two test statistics proposed in this paper in other classes of dynamic models.

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APPENDIX

Proof. (Theorem 1). Under the null hypothesis, the same argument applied in the proof of lemma 1 in Monti (1994) holds so that

$$\hat{\pi}_k = \hat{\rho}_k + O_p(n^{-1}), \tag{A1}$$

which implies that $\hat{\pi}_k = \hat{\rho}_k + o_p(n^{-\delta})$ for $1/2 < \delta < 1$. We start by showing that $\hat{z}_{1k} = z_{1k} + o_p(n^{-1/2})$. Under $\mathcal{H}_0$, there exists $n_0 > 0$ such that, for $n > n_0$, $\hat{\rho}_k$ is contained in a compact subinterval of $[-1, 1]$, for example, $U = [-M, M]$, for $0 < M < 1$, with probability tending to 1. For $n > n_0$ and for any $a \in \mathbb{R}$ with $a \neq 0$,

$$\exp\left\{\frac{\sqrt{n}}{2}\left[\log(1 \pm a\hat{\pi}_k) - \log(1 \pm a\hat{\rho}_k)\right]\right\} = \left(\frac{1 \pm a\hat{\pi}_k}{1 \pm a\hat{\rho}_k}\right)^{\frac{\sqrt{n}}{2}} = \left(1 + \frac{o_p(n^{-\delta})}{1 \pm a\hat{\rho}_k}\right)^{\frac{\sqrt{n}}{2}} \xrightarrow{p} 1,$$

as n tends to infinity, since $\delta > 1/2$. Upon applying the logarithm function to the above expression, the result follows from the continuous mapping theorem, with $a = 1$. Using Slutsky's theorem, we conclude that (Kendall & Stuart, 1977, p. 419)

$$\sqrt{(n-k-3)}\hat{z}_{1k} = \sqrt{(n-k-3)}\left(z_{1k} + o_p(n^{-1/2})\right) \xrightarrow{d} \mathcal{N}(0, 1).$$

Hence, Q_1 and Q_{KW1} are asymptotically equivalent, and $Q_1 \xrightarrow{d} \chi^2_m$.

We shall show that $\hat{z}_{4k} = z_{4k} + o_p(n^{-\delta})$. By using the integral representation $\sin^{-1}(x) = \int_0^x (1 - z^2)^{-\frac{1}{2}} dz$, for any $a \neq 0$, we obtain

$$\begin{aligned} \sin^{-1}(a\hat{\pi}_k) = \sin^{-1}\left(a\hat{\rho}_k + o_p(n^{-\delta})\right) &= \int_0^{a\hat{\rho}_k} \frac{1}{\sqrt{1-z^2}} dz + \int_{a\hat{\rho}_k}^{a\hat{\rho}_k + o_p(n^{-\delta})} \frac{1}{\sqrt{1-z^2}} dz \\ &= \sin^{-1}(a\hat{\rho}_k) + R_n. \end{aligned}$$

For n sufficiently large, $|\hat{\rho}_k + o_p(n^{-\delta})| < M + \varepsilon < 1$, for some $0 < \varepsilon < 1 - M$, so that

$$0 \le R_n = \int_{a\hat{\rho}_k}^{a\hat{\rho}_k + o_p(n^{-\delta})} \frac{1}{\sqrt{1-z^2}} dz \le \frac{1}{\sqrt{1-(M+\varepsilon)^2}} \left|o_p(n^{-\delta})\right| = o_p(n^{-\delta}),$$

and the result follows with $a = 1$. Using Slutsky's theorem, it follows that $\hat{z}_{4k}$ and z_{4k} are asymptotic equivalent, and so are Q_4 and Q_{KW4}. Hence, by Kwan and Sim (1996b), $Q_4 \xrightarrow{d} \chi^2_m$ follows. □

Proof. (Theorem 2). Let us assume for the moment that no covariates are present in the model. We start by showing that $Q_1 \xrightarrow{d} \chi^2_{m-p-q}$. Let $\boldsymbol{\rho}_{(k)} = (\rho_1, \ldots, \rho_k)'$ and $\hat{\boldsymbol{\rho}}_{(k)} = (\hat{\rho}_1, \ldots, \hat{\rho}_k)'$ be the true and sample autocorrelation vector up to lag k, respectively. Since $\mathrm{g}(\mu_t) = \mathrm{g}(y_t) - r_t$, then

$$\mathrm{g}(y_t) = \alpha + \sum_{i=1}^{p} \varphi_i \mathrm{g}(y_{t-i}) + \sum_{j=0}^{q} \theta_j r_{t-j} \quad \Longleftrightarrow \quad \Phi(L)\mathrm{g}(y_t) = \alpha + \Theta(L) r_t, \tag{A2}$$

where L is the lag (backshift) operator and $\theta_0 = 1$. By Assumptions A1 and A3, (A2) is invertible at the true value of $\boldsymbol{\theta}$ as well as in U. Hence, for sufficiently large n, (A2) is invertible at the CMLE $\hat{\boldsymbol{\theta}}$. In view of this, following McLeod (1978), under Assumptions A1 through A3, there exists an idempotent matrix Q of rank $p + q$ such that $\hat{\boldsymbol{\rho}}_{(k)} = (I_k - Q)\boldsymbol{\rho}_{(k)} + O_p(n^{-1})$, where I_k is the k-dimensional identity matrix. Under the null hypothesis, (A1) is still valid, so that by letting $\hat{\boldsymbol{\pi}}_{(k)} = (\hat{\pi}_1, \ldots, \hat{\pi}_k)'$ and following the same steps as in the proof of Theorem 1, we establish that

$$\hat{\boldsymbol{\pi}}_{(k)} = (I_k - Q)\hat{\boldsymbol{\rho}}_{(k)} + o_p(n^{-\delta}).$$

Let $\hat{\boldsymbol{z}}_{1(m)} = (\hat{z}_{11}, \ldots, \hat{z}_{1m})'$ and $\boldsymbol{z}_{1(m)} = (z_{11}, \ldots, z_{1m})'$. Using the argumentation employed in the proof of Theorem 1 (with a being the appropriate entry of the matrix $I_k - Q$), we conclude that

$$\sqrt{(n-k-3)}\hat{\boldsymbol{z}}_{1(m)} = \sqrt{(n-k-3)}\left(\boldsymbol{z}_{1(m)} + o_p(n^{-1/2})\right) \xrightarrow{d} \mathcal{N}_m(\boldsymbol{0}, I_m - Q),$$

where $\mathcal{N}_m(\boldsymbol{0}, \Sigma)$ denotes the m-variate normal distribution with mean vector $\boldsymbol{0} = (0, \ldots, 0)' \in \mathbb{R}^m$ and covariance matrix Σ. Finally, since $I_m - Q$ is an idempotent matrix of rank $p+q$, it follows that

$$Q_1 = (n-k-3)\hat{\boldsymbol{z}}_{1(m)}{}'\hat{\boldsymbol{z}}_{1(m)} \xrightarrow{d} \chi^2_{m-p-q}.$$

Next, we shall prove that $Q_4 \xrightarrow{d} \chi^2_{m-p-q}$. Again, under $\mathcal{H}_0$, (A1) holds, and under Assumptions A1 through A3, the representation in (A2) is valid. Let $\hat{\boldsymbol{z}}_{4(m)} = (\hat{z}_{41}, \ldots, \hat{z}_{4m})'$ and $\boldsymbol{z}_{4(m)} = (z_{41}, \ldots, z_{4m})'$. By making use of the argumentation employed in the proof of Theorem 1 with the appropriate coordinate a, we conclude that

$$\left(\frac{n-k}{n-k-1}\right)\hat{\boldsymbol{z}}_{4(m)} \xrightarrow{d} \mathcal{N}_m(\boldsymbol{0}, I_m - Q),$$

and the result follows. If covariates are present in the model, since $r_t = g(y_t) - g(\mu_t)$, (2) implies that $S_t = g(y_t) - \boldsymbol{x}_t'\boldsymbol{\beta}$ satisfies the ARMA(p, q) difference equations, that is,

$$S_t = \alpha + \sum_{i=1}^{p}\varphi_i S_{t-i} + \sum_{j=0}^{q}\theta_j r_{t-j}, \tag{A3}$$

with $\theta_0 = 1$. The result follows from the arguments outlined in section 3 of Pierce (1972), in light of Box and Pierce (1970) and McLeod (1978). □

Proof. (Corollary 1). The asymptotic null distribution of Q_{LB} follows from using the reasoning presented in Section 3 of Ljung and Box (1978). Finally, the asymptotic null distribution of Q_{M} follows from the arguments used in lemma 1 of Monti (1994). □